

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

The OptiCrop: Smart Agricultural Production Optimization Engine uses machine learning techniques to analyze soil nutrients (Nitrogen, Phosphorous, Potassium) and environmental conditions such as temperature, humidity, pH, and rainfall to recommend the most suitable crop for cultivation. The objective is to help farmers improve crop productivity, optimize resource utilization, reduce farming risks, and promote sustainable agricultural practices through data-driven recommendations.

I am (A farmer/agricultural user)	I'm trying to (Select the most suitable crop for my land based on soil and weather conditions.)	But (I don't know which crop will provide the highest yield under my current environmental conditions.)	Because (Crop selection depends on several factors like soil nutrients, pH, rainfall, temperature, and humidity, which are difficult to analyze manually.)	Which makes me feel (Confused and worried about crop failure and financial loss.)
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Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer planning to cultivate crops.	Identify the most suitable crop for my field to maximize productivity.	I am unsure which crop is best suited to the current soil and environmental conditions.	Crop growth depends on multiple factors such as Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall, making manual decision-making difficult. .	Concerned about poor crop yield, resource wastage, and financial loss.